

## Research Analysis

### **Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection**

#### **1. Introduction**

This Research Analysis examines the empirical results produced by the implemented **Reliability-Aware Deployment Rank Aggregation (RDRA)** framework. Whereas the preceding Research Implementation demonstrated how the proposed method was operationalised as a functioning Python/Google Colab research artefact, the present analysis focuses on interpreting the experimental evidence, identifying relationships within the results, evaluating the research propositions, and determining the extent to which reliability-aware aggregation addresses the Main Research Question.

The research investigates deployment-oriented predictive-maintenance model selection where multiple criteria can support conflicting candidate rankings and where those rankings may differ in empirical reproducibility. The Main Research Question is:

**How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable empirical evidence?**

The analysis therefore examines three related questions:

- **VQ1 - Criterion Uncertainty:** Does primary RDRA produce a more consistent consensus when the available deployment criteria change?
- **VQ2 - Data-Quality Uncertainty:** Does a reliability-aware ranking respond more meaningfully when the deployment evidence deteriorates?
- **VQ3 - Preference Uncertainty:** Does the resulting ranked shortlist provide additional downstream decision usefulness when stakeholder priorities vary?

The purpose is not to establish that RDRA must outperform Equal Borda in every situation. Rather, the analysis investigates under which forms of deployment uncertainty the additional empirical reliability mechanism provides measurable value, and where the current evidence defines limits on that contribution.

## 2. Data Capture

### 2.1 Data Sources

The research uses two predictive-maintenance datasets with different analytical roles.

The **AI4I 2020 Predictive Maintenance Dataset** [8] is the primary source of experimental evidence. It supports binary machine-failure prediction and provides the basis for the complete RDRA analysis. The implemented experiment evaluates 18 candidate configurations derived from two model families, three imbalance-handling strategies and three calibration strategies ( $2 \times 3 \times 3 = 18$ ). Each configuration is evaluated across  $R = 20$  experimental repetitions.

AI4I provides the evidence required to construct the five deployment dimensions used by RDRA:

1. Predictive Quality
2. Robustness
3. Probability Reliability
4. Computational Efficiency
5. Operational Impact

The second source is **NASA C-MAPSS FD001** [6], [20], [21]. C-MAPSS represents a different predictive-maintenance task based on sequential Remaining Useful Life prediction. It is analysed using Ridge, Random Forest, XGBoost, LSTM and GRU candidate models.

C-MAPSS is not used as a second full reliability-weighted RDRA replication. Instead, it provides **complementary cross-task evidence** about deployment-ranking responsiveness under controlled sequential sensor degradation.

### 2.2 AI4I Experimental Data Capture

The AI4I evidence was generated through repeated evaluation of the 18 candidate configurations. The repeated design was necessary because the central RDRA mechanism depends on determining whether criterion-specific candidate rankings are reproducibly supported.

For every experimental repetition, evidence was retained for predictive behaviour, calibration, computational requirements, operational error costs and robustness under controlled data-quality corruption.

The resulting evidence can conceptually be represented as:

$$\textit{Repetition} \times \textit{Candidate} \times \textit{Deployment Evidence}$$

Retaining repetition-level evidence was essential. If results had been averaged immediately across the 20 repetitions, the final mean performance of each candidate could still be calculated,

but the variability in **candidate ordering** across repetitions would be lost. Consequently, empirical ranking reproducibility could not be estimated.

Therefore, the repeated design supports two different analytical levels:

1. Candidate-level evidence – final deployment ranking
2. Repetition-level evidence - criterion reliability estimation

### 2.3 Data Captured for the Five Deployment Dimensions

The heterogeneous experimental outputs were organised into five deployment dimensions

**Table 2.1 Deployment evidence captured for analysis**

| Deployment dimension     | Main evidence                    | Analytical interpretation                                |
|--------------------------|----------------------------------|----------------------------------------------------------|
| Predictive Quality       | F1-score, PR-AUC                 | Ability to predict failure effectively                   |
| Robustness               | Corruption-performance retention | Resistance to degraded sensor evidence                   |
| Probability Reliability  | Brier score, ECE                 | Reliability of predicted probabilities                   |
| Computational Efficiency | Inference latency, model size    | Practical computational deployment requirements          |
| Operational Impact       | Asymmetric prediction cost       | Consequences of false-negative and false-positive errors |

The dimensions intentionally capture different deployment perspectives. Consequently, disagreement between them is analytically meaningful rather than being treated as experimental noise. For example, strong predictive quality does not imply strong computational efficiency, and good discriminative ability does not guarantee good probability calibration. These properties capture different aspects of deployment suitability and must be evaluated independently.

This multi-dimensional structure forms the empirical basis for investigating whether a deployment consensus should treat every criterion ranking as equally reliable.

### 2.4 Controlled Data-Quality Capture for VQ2

The VQ2 analysis uses 16 controlled AI4I corruption scenarios from four corruption families:

- Gaussian noise
- Missing Completely At Random
- outlier contamination
- static sensor offset

For each corruption scenario, two quantities were derived:

$\Delta E$  = deployment evidence-change magnitude

and:

$\Delta R$  = resulting ranking-change magnitude

This design allows the analysis to investigate whether larger changes in the deployment evidence are associated with correspondingly larger changes in model ranking.

A fixed clean-data normalisation reference was retained when evaluating corrupted evidence. This prevents scenario-specific rescaling from hiding deterioration by merely renormalising degraded candidates relative to one another.

## 2.5 Stakeholder-Preference Data Capture for VQ3

The preference-uncertainty experiment generated 5,000 matched stakeholder-preference vectors using a Dirichlet distribution.

Each vector represents a possible distribution of stakeholder priorities across the deployment objectives and satisfies:

$$w_k \geq 0$$

and:

$$\sum_k w_k = 1.$$

The same 5,000 preference vectors were applied to the candidate shortlists produced by RDRA and Equal Borda.

Using matched scenarios is important because both aggregation methods are exposed to the same downstream decision conditions. Differences in resulting regret therefore cannot be attributed to different preference samples.

The analysis focuses primarily on **Top-3 shortlists**, reflecting a practical decision-support setting in which a ranking system reduces the model-candidate space rather than automatically replacing the final human or organisational decision.

## 2.6 C-MAPSS Data Capture

C-MAPSS FD001 provides sequential degradation evidence for five candidate-model families:

- Ridge
- Random Forest

- XGBoost
- LSTM
- GRU

Three deployment dimensions are currently available:

- Predictive Quality
- Robustness
- Computational Efficiency

The analysis also includes controlled temporal degradation conditions, including Gaussian noise, gradual offset drift, missing blocks, outliers and sensor freezing.

Because this branch does not contain the same repeated five-dimension structure as AI4I, it is used to investigate whether deployment rankings respond to data-quality degradation in another predictive-maintenance task rather than to estimate an independent full RDRA reliability-weight vector.

## **2.7 Reliability, Validity and Bias Controls in Data Capture**

Several controls were applied to improve the quality and interpretability of the captured evidence.

First, all candidate configurations were evaluated through consistent experimental procedures rather than being compared using unrelated evidence-generation pipelines.

Second, repeated AI4I experiments were retained to reduce dependence on one experimental run and to permit direct estimation of ranking reproducibility.

Third, tie-aware ranking was used so that candidates with equivalent evidence were not arbitrarily ordered.

Fourth, the project uses a fixed random state:

$$\text{RNG\_SEED} = 20260811,$$

which improves reproducibility of randomised analytical procedures.

Fifth, VQ1 comparisons are paired because the same criterion subsets are evaluated under RDRA and Equal Borda.

Sixth, VQ2 uses matched corruption scenarios and a matched-bootstrap comparison rather than comparing significance decisions from unrelated tests.

Seventh, VQ3 uses the same 5,000 stakeholder-preference vectors for both aggregation methods.

Finally, the exploratory reliability-influence parameter was selected using development-only VQ1 evidence and locked before VQ2 and VQ3 evaluation. This separation reduces the risk of validation leakage.

## 2.8 Data-Capture Limitations

The captured evidence also has limitations.

The primary complete RDRA analysis is based on one AI4I experimental environment. Consequently, the empirical reliability weights should be interpreted as **protocol-specific estimates of ranking reproducibility**, not universal values applicable to every predictive-maintenance system.

The five deployment dimensions are also not exhaustive. Other deployment contexts could require additional considerations such as maintainability, energy consumption, cybersecurity, regulatory requirements or explainability.

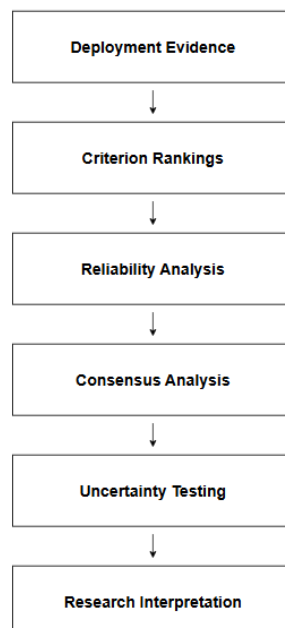
Furthermore, C-MAPSS currently contributes only three deployment dimensions and does not provide an independent five-dimension reliability-weight replication.

These limitations establish the boundary of the present analysis rather than invalidating the experimental evidence.

## 3. Data Analysis Methodology

### 3.1 Analytical Strategy

The analysis follows six stages:



### Figure 3.1 Analytical Sequence of the Research Data-Analysis Methodology

The objective is not merely to describe final rankings but to analyse:

- whether criterion evidence differs in reproducibility
- whether reliability-aware aggregation changes ranking behaviour
- whether those changes improve consistency
- whether the ranking responds appropriately to evidence deterioration
- whether ranking-level improvements produce downstream decision benefits

#### 3.2 Hypothesis and Validation Framework

For analytical clarity, the validation questions can be expressed as testable research propositions.

##### H1: Criterion-Uncertainty Consistency

**Primary RDRA produces greater ranking and shortlist consistency than Equal Borda when the available deployment criteria change.**

Evidence is evaluated through Kendall agreement, Spearman correlation, Top-3 overlap and mean absolute rank displacement.

##### H2: Data-Quality Responsiveness

**The locked exploratory refined RDRA exhibits a stronger association between deployment-evidence change and ranking change than Equal Borda.**

This is tested through the direct difference:

$$\Delta\rho = \rho_{\text{Refined RDRA}} - \rho_{\text{Borda}}.$$

##### H3: Downstream Preference Usefulness

**RDRA provides greater Top-3 downstream decision usefulness than Equal Borda under varying stakeholder preferences.**

This is evaluated using matched preference scenarios and regret/Pareto-based evaluation.

The analysis determines whether the available experimental evidence supports or does not support each proposition.

#### 3.3 Descriptive and Inferential Analysis

Both descriptive and inferential methods are used.

Descriptive analysis includes:

- mean Kendall agreement

- mean Spearman correlation
- Top-3 overlap
- absolute rank displacement
- Top-1 survival
- reliability-weight distributions
- corruption-response trends
- downstream regret

Inferential analysis includes:

- paired statistical comparisons
- 95% confidence intervals
- Holm correction for multiple VQ1 comparisons
- Spearman correlation
- matched bootstrap resampling

This combination is crucial because descriptive differences indicate the direction and magnitude of observed effects, whereas inferential procedures help determine whether the differences are sufficiently consistent across the available scenarios to support stronger conclusions.

4. Results and In-Depth Analysis

4.1 Criterion-Level Empirical Reliability

The first important result is that the five deployment criteria do not demonstrate equal ranking reproducibility.

|   | Dimension   | Mean_Kendall_Tau_100 | Stability_Weight_Raw | N_Repetitions | Reliability_Weight |
|---|-------------|----------------------|----------------------|---------------|--------------------|
| 0 | Predictive  | 0.651369             | 0.651369             | 20            | 0.222560           |
| 1 | Robustness  | 0.376062             | 0.380964             | 20            | 0.130168           |
| 2 | Reliability | 0.788889             | 0.788889             | 20            | 0.269549           |
| 3 | Efficiency  | 0.818301             | 0.818301             | 20            | 0.279598           |
| 4 | Operational | 0.258432             | 0.287183             | 20            | 0.098125           |

Figure 4.1 Empirical criterion-ranking stability estimates and corresponding reliability weights

Table 4.1 Empirical AI4I reliability estimates

| Dimension                | Positive stability | Reliability weight |
|--------------------------|--------------------|--------------------|
| Predictive Quality       | 0.651369           | 0.222560           |
| Robustness               | 0.380964           | 0.130168           |
| Probability Reliability  | 0.788889           | 0.269549           |
| Computational Efficiency | 0.818301           | 0.279598           |
| Operational Impact       | 0.287183           | 0.098125           |

The largest empirical reliability weight is associated with Computational Efficiency  $w_{\text{Efficiency}} = 0.279598$ , followed closely by Probability Reliability  $w_{\text{Reliability}} = 0.269549$ .

Predictive Quality receives 0.222560, while Robustness and Operational Impact receive substantially smaller values 0.130168 and 0.098125, respectively.

The pattern provides empirical support for the motivation behind RDRA: an equal-weight assumption of  $w_j = 0.20$  does not reflect the observed reproducibility of the five criterion rankings.

However, this result should not be interpreted as evidence that Computational Efficiency is inherently more important than Operational Impact. The weights measure reproducibility of candidate ordering, not stakeholder importance.

A more appropriate interpretation is:

higher  $w_j \rightarrow$  greater empirical confidence in criterion  $j$ 's ordering.

The relatively high stability of Efficiency suggests that candidates retained a similar ordering across repeated experiments for latency/model-size evidence. By contrast, the lower Operational Impact weight indicates greater repetition-to-repetition variability in the candidate ordering generated from operational-cost evidence.

This difference is important because RDRA responds to uncertainty in the ranking itself, rather than simply uncertainty in individual metric values.

## 4.2 Interpretation of Signed and Positive Stability

For Predictive Quality, Probability Reliability and Computational Efficiency, the signed mean Kendall agreement is identical to the positive stability value.

However, for Robustness and Operational Impact:

Positive Stability > Mean Signed Kendall.

For example, Operational signed mean = 0.258432, whereas  $s_{\text{Operational}} = 0.287183$ .

This difference occurs because negative repetition-level Kendall agreements are clipped to zero before calculating the positive-stability measure.

The distinction reveals that at least some repetitions produce ordering disagreement rather than merely weak positive agreement.

This is analytically meaningful. Negative rank agreement suggests that the candidate ordering under an individual repetition can partially reverse relative to the reference evidence. RDRA does

not treat this as evidence that a criterion should become directionally negative; instead, it treats the disagreement as lack of positive reliability evidence.

This behaviour maintains  $w_j \geq 0$  and avoids transforming uncertainty into criterion inversion.

### 4.3 VQ1 - Criterion-Uncertainty Results

Across the 25 criterion-subset perturbations, primary RDRA demonstrates stronger descriptive consistency on several major ranking measures.

|   | Method             | Mean_Kendall | Mean_Spearman | Mean_Top3_Overlap | Mean_Absolute_Rank_Displacement | Tie_Aware_Winner_Survival | Mean_Top1_Set_Overlap | N_Perturbations |
|---|--------------------|--------------|---------------|-------------------|---------------------------------|---------------------------|-----------------------|-----------------|
| 0 | Equal_Borda        | 0.677306     | 0.799260      | 0.720000          | 2.257778                        | 0.32                      | 0.20                  | 25              |
| 1 | Existing_Composite | 0.624052     | 0.741754      | 0.746667          | 2.448889                        | 0.36                      | 0.36                  | 25              |
| 2 | RDRA               | 0.746405     | 0.862456      | 0.813333          | 1.773333                        | 0.36                      | 0.36                  | 25              |

**Figure 4.2 VQ1 comparison of Equal Borda, Existing Composite and RDRA across 25 criterion-subset perturbations**

**Table 4.2 VQ1 descriptive results**

| Method              | Kendall         | Spearman        | Top-3 overlap   | Rank displacement |
|---------------------|-----------------|-----------------|-----------------|-------------------|
| Existing Composite  | 0.624052        | 0.741754        | 0.746667        | 2.448889          |
| Equal Borda         | 0.677306        | 0.799260        | 0.720000        | 2.257778          |
| <b>Primary RDRA</b> | <b>0.746405</b> | <b>0.862456</b> | <b>0.813333</b> | <b>1.773333</b>   |

Primary RDRA increases mean Kendall agreement by approximately:

$$0.746405 - 0.677306 = 0.069099.$$

This indicates stronger preservation of pairwise candidate ordering when criteria are removed.

The increase in Top-3 overlap is:

$$0.813333 - 0.720000 = 0.093333.$$

This is particularly relevant operationally because deployment ranking may often be used to create a shortlist rather than to enforce one exact winner.

RDRA also reduces mean absolute rank displacement by approximately:

$$2.257778 - 1.773333 = 0.484445$$

positions.

Therefore, these measures show that the improvement is not confined to one definition of consistency. RDRA demonstrates stronger behaviour in pairwise ordering, shortlist preservation and overall movement magnitude.

#### 4.4 Statistical Interpretation of VQ1

The paired VQ1 results strengthen the descriptive findings.

|   | Metric            | Mean improvement | 95% CI          | Holm-adjusted paired t-test | Effect   |
|---|-------------------|------------------|-----------------|-----------------------------|----------|
| 0 | Kendall           | +0.069           | [0.012, 0.126]  | 0.039                       | Moderate |
| 1 | Spearman          | +0.063           | [-0.007, 0.134] | 0.077                       | Small    |
| 2 | Top-3 overlap     | +0.093           | [0.030, 0.156]  | 0.022                       | Moderate |
| 3 | Rank displacement | +0.484 better    | [0.146, 0.822]  | 0.022                       | Moderate |

Figure 4.3 VQ1 paired statistical-validation results

Table 4.3 Paired primary-RDRA versus Equal-Borda results

| Measure           | Improvement        | 95% CI          | Holm-adjusted $p$ |
|-------------------|--------------------|-----------------|-------------------|
| Kendall           | +0.069             | [0.012, 0.126]  | 0.039             |
| Spearman          | +0.063             | [-0.007, 0.134] | 0.077             |
| Top-3 overlap     | +0.093             | [0.030, 0.156]  | 0.022             |
| Rank displacement | 0.484 ranks better | [0.146, 0.822]  | 0.022             |

Three measures remain statistically supported after Holm correction:

- Kendall agreement
- Top-3 overlap
- rank displacement

The Spearman result is directionally positive but its confidence interval crosses zero and  $p_{\text{Holm}} = 0.077$ . Consequently, the evidence does **not** justify describing every VQ1 metric as statistically significant.

#### H1 assessment

The evidence therefore **supports H1 on several major consistency measures**, but not universally across every evaluated metric.

This is stronger than a purely descriptive claim because the observed improvement persists across matched criterion perturbations and remains statistically supported after adjustment for multiple testing.

#### 4.5 An Important VQ1 Finding: Shortlist Stability Is Stronger Than Winner Stability

One interesting result is that RDRA's clearest benefit occurs at the broader ranking and Top-3 levels rather than exact winner preservation.

Tie-aware winner survival for primary RDRA is 0.36, which is the same as the Existing Composite and only modestly above Equal Borda 0.32. By contrast, primary RDRA's Top-3 overlap reaches 0.813333. This suggests that reliability-aware aggregation is more effective at stabilising **the high-quality candidate region** than guaranteeing one immutable winner.

That pattern is practically reasonable. When several candidate configurations have similar deployment evidence, small criterion changes can legitimately alter which candidate occupies rank 1 while leaving the set of strongest deployment choices largely unchanged.

This indicates that RDRA may be particularly appropriate as a shortlisting mechanism rather than as a mechanism that claims one universally superior model.

#### 4.6 Reliability-Weight Sensitivity Analysis

The 1,000 reliability-weight perturbations provide evidence about whether the primary RDRA ranking is excessively dependent on one precise estimated weight vector.

| RDRA weight-estimation sensitivity |        |                 |              |                |                   |                        |                         |                       |
|------------------------------------|--------|-----------------|--------------|----------------|-------------------|------------------------|-------------------------|-----------------------|
|                                    | N_Runs | Perturbation_SD | Mean_Kendall | Median_Kendall | Mean_Top3_Overlap | Mean_Rank_Displacement | Tie_Aware_Top1_Survival | Mean_Top1_Set_Overlap |
| 0                                  | 1000   | 0.05            | 0.964261     | 0.973856       | 1.0               | 0.296444               | 0.699                   | 0.699                 |

**Figure 4.4 Sensitivity of RDRA consensus to reliability-weight perturbation**

The results were:

- Mean Kendall = 0.964201
- Median Kendall = 0.973856
- Mean Top-3 overlap = 1.000000
- Mean absolute rank displacement = 0.296444
- Tie-aware Top-1 survival = 0.699
- Mean Top-1 set overlap = 0.699

These values indicate very high overall ranking stability. Most importantly, Mean Top-3 overlap = 1.0 shows that the shortlist is preserved under the tested moderate weight perturbations. However, exact Top-1 survival is 0.699. The contrast between perfect Top-3 preservation and lower winner survival again shows that candidate competition is concentrated near the top of the ranking.

Therefore, the result suggests RDRA shortlist robustness > RDRA exact-winner robustness. This is an important qualification because it prevents stability findings from being interpreted as evidence of an invariant single best candidate.

#### 4.7 VQ2 - Data-Quality Responsiveness

The VQ2 analysis investigates whether ranking changes correspond meaningfully to changes in deployment evidence.

|   | Method        | Spearman_Rho         | P_Value_Two_Sided           | Bootstrap_CI_Lower   | Bootstrap_CI_Upper   | N_Scenarios | N_Bootstrap_Valid |
|---|---------------|----------------------|-----------------------------|----------------------|----------------------|-------------|-------------------|
| 0 | Refined RDRA  | 0.678041             | 0.003892                    | 0.180088             | 0.937348             | 16          | 10000             |
| 1 | Equal Borda   | 0.347570             | 0.187141                    | -0.235391            | 0.825839             | 16          | 10000             |
|   | RDRA_Spearman | Equal_Borda_Spearman | Difference_RDRA_minus_Borda | Bootstrap_CI95_Lower | Bootstrap_CI95_Upper | N_Scenarios |                   |
| 0 | 0.678041      | 0.34757              | 0.330471                    | 0.046026             | 0.702235             | 16          |                   |

**Figure 4.5 RDRA versus Equal Borda responsiveness comparison**

For the locked exploratory refined RDRA,  $\rho_{\text{Refined RDRA}} = 0.678041$  with  $p = 0.003892$ .

For Equal Borda,  $\rho_{\text{Borda}} = 0.347570$  with  $p = 0.187141$ .

Therefore, the difference is  $\Delta\rho = \rho_{\text{Refined RDRA}} - \rho_{\text{Borda}} = 0.330471$ .

The result indicates that evidence deterioration is substantially more closely associated with ranking movement under the locked refined method.

This is important because a ranking mechanism should not only resist unnecessary changes. It should also react when the evidence that supports candidate deployment suitability changes substantially.

Thus, VQ2 provides a complementary perspective to VQ1:

- VQ1 = avoid unjustified change
- VQ2 = respond to justified change

#### 4.8 Direct Statistical Comparison of VQ2

The strongest VQ2 evidence is not the difference between  $p_{\text{Refined RDRA}} < 0.05$  and  $p_{\text{Borda}} > 0.05$ .

Those two significance decisions alone do not establish that the correlations differ.

The direct matched-bootstrap comparison addresses the actual difference:

$$\Delta\rho = \rho_{\text{Refined RDRA}} - \rho_{\text{Borda}}$$

Using 10,000 matched bootstrap resamples, the 95% confidence interval is [0.046026, 0.702235]. Because the interval remains entirely above zero, the evidence

supports the conclusion that refined RDRA is more responsive than Equal Borda under the tested corruption scenarios.

## H2 assessment

Hence, the evidence **supports H2** for the locked exploratory  $\gamma = 3$  RDRA configuration.

This finding must remain specifically attributed to the locked exploratory refinement. It should not be transferred automatically to primary  $\gamma = 1$  RDRA.

## 4.9 Relationship between VQ1 and VQ2

Therefore, VQ1 and VQ2 reveal a particularly important relationship.

A ranking method could theoretically achieve strong VQ1 results simply by being reluctant to change. However, such a method would be undesirable if it failed to react when deployment evidence genuinely deteriorated.

The present results instead show evidence for two complementary behaviours:

- Primary RDRA - greater criterion-set consistency
- Locked refined RDRA - greater evidence responsiveness

Therefore, the reliability-aware approach is not supported solely because it creates a more static ranking.

The results instead suggest that reliability-aware aggregation can distinguish, at least under the tested conditions, between structural uncertainty in available criteria and meaningful degradation of underlying evidence.

This distinction is central to the claim that the resulting consensus is more defensible, rather than merely more stable.

## 4.10 VQ3 - Preference-Uncertainty Results

|   | Method             | Top3_Feasible_Shortlist                           | Mean_Shortlist_Regret | P95_Shortlist_Regret | CVar95_Shortlist_Regret | Worst_Case_Shortlist_Regret | Zero_Regret_Frequency |
|---|--------------------|---------------------------------------------------|-----------------------|----------------------|-------------------------|-----------------------------|-----------------------|
| 0 | Equal_Borda        | XGBoost   class_weight   isotonic   XGBoost   ... | 0.002069              | 0.016802             | 0.038217                | 0.079113                    | 0.9316                |
| 1 | RDRA               | XGBoost   class_weight   isotonic   XGBoost   ... | 0.002069              | 0.016802             | 0.038217                | 0.079113                    | 0.9316                |
| 2 | Existing_Composite | XGBoost   class_weight   platt   XGBoost   cla... | 0.003082              | 0.019930             | 0.057028                | 0.139577                    | 0.9260                |

**Figure 4.6 VQ3 ranking shortlist decision usefulness**

**Table 4.4 VQ3 Top-3 downstream decision-usefulness results across 5,000 matched preference scenarios**

| Method             | Mean regret     | Worst-case regret | Zero-regret frequency |
|--------------------|-----------------|-------------------|-----------------------|
| Equal Borda        | <b>0.002069</b> | <b>0.079113</b>   | <b>0.9316</b>         |
| RDRA               | <b>0.002069</b> | <b>0.079113</b>   | <b>0.9316</b>         |
| Existing Composite | 0.003082        | 0.139577          | 0.9260                |

The VQ3 analysis provides a different outcome.

Across 5,000 matched stakeholder-preference scenarios, RDRA and Equal Borda show broadly comparable Top-3 downstream decision usefulness. Therefore, the current evidence does not demonstrate that RDRA Top-3 downstream advantage exceeds the Equal Borda Top-3 downstream advantage.

Accordingly, H3 is not supported as a superiority hypothesis under the tested preference scenarios. RDRA maintained descriptively comparable Top-3 downstream decision usefulness to Equal Borda, but no additional RDRA advantage was demonstrated. Because no equivalence or non-inferiority test was conducted, this result should not be interpreted as formal statistical equivalence.

### **H3 assessment**

The VQ3 result suggests that the benefit of RDRA lies primarily in improving the behaviour and defensibility of the ranking process, rather than necessarily producing a better downstream Top-3 shortlist under every stakeholder-preference configuration.

The conclusion is RDRA maintained broadly comparable Top-3 downstream decision usefulness to Equal Borda under the tested preference scenarios, but no additional RDRA advantage was demonstrated.

#### **4.11 Why the VQ3 Result Matters**

The neutral VQ3 result is scientifically important because it prevents the analysis from interpreting every positive ranking-level result as evidence of universal superiority.

The finding demonstrates that better ranking consistency does not imply better downstream decision usefulness. Similarly, stronger evidence responsiveness does not guarantee superior performance under every stakeholder-preference scenario.

Under the tested 5,000 matched preference scenarios, RDRA and Equal Borda produced broadly comparable Top-3 downstream decision usefulness. This suggests that the principal benefit of

RDRA may lie in improving the robustness and defensibility of the ranking process, rather than necessarily producing a better Top-3 shortlist under every possible stakeholder-priority configuration.

One plausible explanation is that RDRA and Equal Borda generate sufficiently similar high-quality Top-3 candidate sets that stakeholder optimisation within those shortlists has access to broadly comparable alternatives. Consequently, changes in the internal ordering of candidates may not substantially affect the quality of the available downstream choices.

Another possibility is that reliability weighting primarily improves ordering robustness and evidence responsiveness, whereas downstream preference regret depends more strongly on whether candidates representing objective trade-offs are included in the shortlist. In this case, improved ranking behaviour would not necessarily reduce regret if both methods already retain similar high-value alternatives.

These interpretations should be treated as plausible explanations rather than established causal findings, because the current experiments were not designed to isolate the causal mechanism behind the neutral VQ3 result. Nevertheless, the finding identifies an important direction for future research: examining when improvements in ranking reliability translate into measurable downstream decision benefits.

4.12 Joint Preference and Corruption Findings

|   | Variant                                 | Mean_Joint_Regret | P95_Joint_Regret | Worst_Case_Joint_Regret | Zero_Regret_Frequency | Joint_Robust_Regret_Rank |
|---|-----------------------------------------|-------------------|------------------|-------------------------|-----------------------|--------------------------|
| 3 | XGBoost   class_weight   platt          | 0.123966          | 0.560538         | 0.969285                | 0.089765              | 1                        |
| 2 | XGBoost   class_weight   isotonic       | 0.084179          | 0.534891         | 0.971801                | 0.686618              | 2                        |
| 0 | Random Forest   class_weight   isotonic | 0.533107          | 0.981915         | 0.998236                | 0.043500              | 3                        |
| 1 | Random Forest   class_weight   platt    | 0.530673          | 1.000000         | 1.000000                | 0.180118              | 4                        |

Figure 4.7 VQ3 joint preference-plus-corruption regret analysis

The joint preference-plus-corruption analysis provides candidate-level evidence under simultaneous uncertainty.

Several XGBoost configurations occupy stronger robust-regret positions than the evaluated Random Forest configurations.

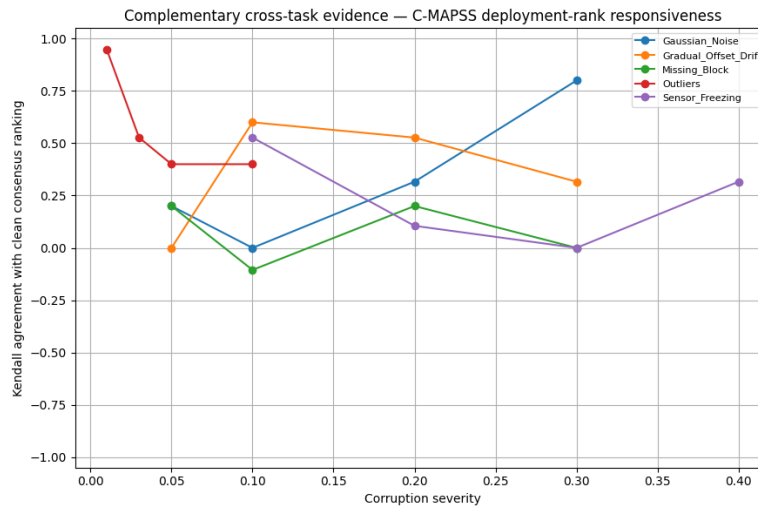
This indicates that certain candidate families remain attractive across a wider combination of stakeholder preferences and degraded evidence conditions.

However, the result does not directly demonstrate RDRA superiority because the joint analysis concerns candidate robustness rather than a matched comparison between the RDRA and Equal Borda aggregation mechanisms.

Therefore, the conclusion is the joint analysis identifies candidate configurations that remain comparatively defensible under simultaneous preference and data-quality uncertainty.

#### 4.13 Complementary C-MAPSS Findings

The C-MAPSS results broaden the analysis beyond AI4I classification.



**Figure 4.8 C-MAPSS complementary cross-task responsiveness**

The sequential RUL experiment shows that candidate ranking agreement with the clean-data reference changes across:

- Gaussian noise
- gradual offset drift
- missing blocks
- outliers
- sensor freezing.

The pattern is not identical across corruption families or severity levels.

This is meaningful because it shows that deployment-rank responsiveness is not unique to the AI4I classification context.

However, C-MAPSS does not provide the complete repeated five-dimension structure required for a second independent RDRA reliability-weight analysis.

Consequently, the conclusion is C-MAPSS provides complementary cross-task evidence that deployment-oriented rankings can respond differently to controlled data-quality degradation in sequential RUL prediction.

## 5. Cross-Result Analysis and Discussion

The purpose of this section is to integrate the separate empirical findings and determine what they collectively indicate about the proposed Reliability-Aware Deployment Rank Aggregation (RDRA) approach. Rather than interpreting VQ1, VQ2 and VQ3 as isolated experiments, the analysis considers how criterion reproducibility, ranking consistency, sensitivity to reliability weights, responsiveness to degraded evidence and downstream preference usefulness relate to one another.

The results support a differentiated rather than universal conclusion. Reliability-aware aggregation demonstrates measurable benefits for several ranking-level properties under the tested uncertainty conditions, while the downstream preference analysis identifies an important boundary on those benefits. This distinction is consistent with the design of RDRA, which separates empirical confidence in criterion-specific evidence from stakeholder preference or criterion importance.

### 5.1 Overall Relationship among the Findings

The combined results form a coherent but bounded pattern.

First, the empirical reliability analysis demonstrates that the five deployment criteria do not reproduce their candidate rankings with equal consistency. Computational Efficiency and Probability Reliability show comparatively stronger repetition-to-repetition ranking reproducibility, while Robustness and Operational Impact show weaker reproducibility. This directly supports the motivation for examining an alternative to the Equal Borda assumption that every criterion should contribute identical influence.

Second, VQ1 demonstrates that primary RDRA improves several measures of ranking consistency under criterion uncertainty. Compared with Equal Borda, primary RDRA achieved stronger Kendall agreement, greater Top-3 overlap and lower mean absolute rank displacement, with these improvements remaining statistically supported after Holm correction on those measures.

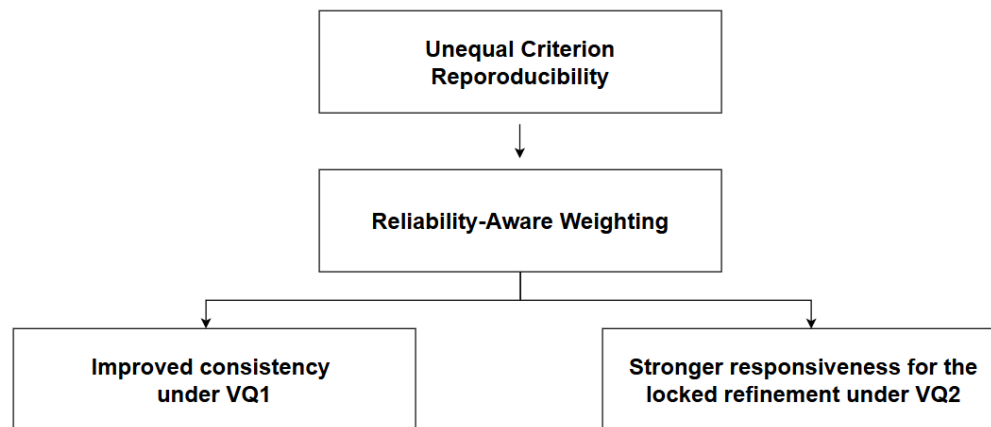
Third, the reliability-weight sensitivity analysis shows that the broader primary RDRA consensus is not excessively dependent on one precisely estimated reliability-weight vector. Across 1,000 moderate perturbations, whole-ranking agreement remained high and the Top-3 shortlist was preserved, although exact Top-1 survival was more sensitive. This indicates robustness of the broader consensus rather than proof that every VQ1 advantage would remain unchanged under every possible alternative weight vector.

Fourth, VQ2 shows that the locked exploratory refinement, which applies stronger reliability differentiation through  $\gamma = 3$ , demonstrated stronger responsiveness to meaningful AI4I

evidence deterioration than Equal Borda under the tested corruption scenarios. The direct matched-bootstrap comparison provides the key inferential support for this result.

Fifth, VQ3 shows that the positive ranking-level findings do not automatically produce an additional downstream Top-3 preference advantage. Across the 5,000 matched stakeholder-preference scenarios, RDRA and Equal Borda were descriptively broadly comparable in downstream shortlist usefulness. The current evidence therefore does not demonstrate an additional RDRA advantage on this specific downstream measure.

Therefore, the overall relationship can be represented as:



**Figure 5.1 Conceptual outcomes of reliability-aware weighting under unequal criterion reproducibility**

while ranking-level improvement is not equal to universal downstream superiority.

This is the central analytical result of the research. The contribution is not that RDRA dominates Equal Borda in every situation, but that empirical ranking reproducibility can provide useful additional information for deployment-oriented aggregation under forms of uncertainty.

## 5.2 Answer to the Main Research Question

The Main Research Question asks:

**How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable empirical evidence?**

The findings indicate that RDRA addresses this problem by introducing an intermediate **evidence-confidence layer** between criterion-specific model evaluation and final consensus formation.

Equal Borda assumes identical influence across all *m*criteria:

$$w_1 = w_2 = \dots = w_m = \frac{1}{m}.$$

With five deployment dimensions:

$$w_j = 0.20.$$

RDRA instead estimates criterion-specific empirical ranking reproducibility:

$$s_j,$$

and converts it into the primary reliability-weight vector:

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k}.$$

Thus, the resulting weight represents the extent to which repeated experimental evidence consistently supports a criterion-specific candidate ordering. A criterion with weaker empirical reproducibility remains represented, but contributes proportionally less to the final consensus. The method consequently moderates criterion influence rather than eliminating uncertain deployment perspectives. This interpretation is consistent with the implemented LOO reliability-estimation and weighting procedure.

The empirical results indicate that this mechanism is associated with several desirable properties:

- greater whole-ranking consistency under criterion uncertainty
- stronger Top-3 shortlist preservation
- lower mean candidate displacement
- high robustness of the broader consensus to moderate reliability-weight perturbation
- for the locked exploratory refinement, stronger responsiveness to meaningful data-quality deterioration

Therefore, the current evidence supports the conclusion that Reliability-aware rank aggregation can produce a more defensible deployment consensus by allowing criterion influence to depend partly on empirical ranking reproducibility rather than assuming identical evidence confidence across all deployment criteria.

### 5.3 Relationship with Existing Literature

The findings are consistent with predictive-maintenance literature showing that practical model selection involves more than predictive accuracy alone. Industrial deployment can require consideration of robustness, calibration, computational requirements and operational consequences in addition to conventional predictive performance [1]–[7].

The current results support this broader deployment perspective because the five operationalised dimensions do not provide identical candidate orderings or identical levels of ranking reproducibility. A candidate that performs strongly according to Predictive Quality does not necessarily occupy the same relative position according to Computational Efficiency, Probability Reliability or Operational Impact.

Research on probability calibration similarly demonstrates that strong discriminatory performance does not necessarily imply trustworthy predicted probabilities [8]–[10]. This supports the decision to preserve Probability Reliability as a separate deployment dimension rather than incorporating it directly into Predictive Quality.

Likewise, research on asymmetric maintenance consequences supports the separate treatment of Operational Impact [11]. False-negative and false-positive errors can impose different practical costs, meaning that predictive performance alone may not adequately characterise deployment suitability.

Multi-criteria decision-making and rank-aggregation research provides established approaches for combining competing criteria and rankings [12]–[19]. However, weighting within conventional MCDM commonly represents criterion importance, preference or decision-maker priorities.

The results indicate that weighting should not be interpreted solely as an expression of criterion importance. Instead, RDRA shows that weights can also encode the empirical reliability of the underlying evidence, two conceptually distinct roles that must not be conflated.

RDRA does not replace preference-based MCDM. Instead, it adds a distinct empirical layer in which criterion influence reflects how reproducibly repeated evidence supports the corresponding candidate ordering.

The resulting framework yields two analytically independent constructs. Empirical reliability quantifies how trustworthy the underlying evidence is, while stakeholder preference encodes what the decision-maker values. These represent different types of influence and should not be conflated.

The VQ3 analysis reinforces the importance of this distinction. Reliability-aware ranking can improve particular ranking properties without eliminating the need for stakeholder-specific downstream decision-making.

## 5.4 Unexpected and Interesting Findings

Several findings provide additional insight beyond the primary validation outcomes.

### 5.4.1 The Most Reproducible Criterion Was Not Predictive Quality

A notable result is that Computational Efficiency, rather than Predictive Quality, produced the largest empirical reliability weight:

$$w_{\text{Efficiency}} = 0.279598.$$

Probability Reliability was close behind:

$$w_{\text{Probability Reliability}} = 0.269549,$$

while Predictive Quality received:

$$w_{\text{Predictive}} = 0.222560.$$

Operational Impact had the smallest weight:

$$w_{\text{Operational}} = 0.098125.$$

These values show that reliability should not be inferred from intuitive perceptions of criterion importance. Computational Efficiency receiving the largest reliability weight does **not** indicate that it is the most important industrial criterion. It means that, under the implemented AI4I experimental protocol, its candidate ordering was reproduced most consistently across the 20 repetitions.

This finding provides direct empirical justification for maintaining the distinction between evidence confidence and decision importance.

### 5.4.2 Shortlist Stability Was Stronger than Exact-Winner Stability

A second interesting result is the contrast between broader shortlist stability and exact Top-1 stability.

In the reliability-weight sensitivity analysis, mean Top-3 overlap remained: 1.000, while tie-aware Top-1 survival was 0.699.

This indicates that the broader high-quality candidate set remained highly stable even when the identity of the exact first-ranked candidate sometimes changed.

The pattern suggests that deployment-oriented rank aggregation may be especially useful as a shortlisting mechanism. When several top candidates have similar deployment evidence, modest changes in weighting may legitimately alter their exact ordering without substantially changing the set of defensible options.

Consequently, shortlist robustness > exact-winner robustness under the tested weight perturbations. This result argues against treating rank 1 as unquestionably superior when the underlying candidates are closely competitive.

#### **5.4.3 Stronger Ranking Behaviour Did Not Guarantee Stronger Downstream Preference Performance**

A third important finding is that positive VQ1 and VQ2 results did not automatically translate into an additional VQ3 downstream advantage.

The VQ3 results show that RDRA and Equal Borda were broadly comparable in Top-3 downstream usefulness across the 5,000 matched stakeholder-preference scenarios. The current evidence therefore does not demonstrate that improved ranking consistency or responsiveness necessarily produces lower downstream regret in every stakeholder setting.

This creates an important analytical distinction between quality of ranking behaviour and utility of the resulting shortlist for a particular stakeholder.

Hence, the neutral VQ3 finding is not evidence that RDRA is ineffective. Instead, it establishes that the demonstrated value of RDRA is concentrated more strongly in ranking defensibility, consistency and responsiveness than in guaranteed improvement on every downstream decision metric.

#### **5.4.4 The Strength of Reliability Influence Appears to Affect Ranking Behaviour**

The fourth finding concerns the exploratory reliability-influence parameter  $\gamma$ .

Primary RDRA uses  $\gamma = 1$ , which corresponds to direct proportional weighting by empirical stability.

The development-selected exploratory configuration uses  $\gamma = 3$ , which increases the relative influence of differences in criterion reproducibility.

The locked  $\gamma = 3$  refinement demonstrated stronger VQ2 responsiveness than Equal Borda under the tested AI4I corruption scenarios. However, this should not be interpreted as evidence that stronger reliability weighting is always preferable.

Importantly,  $\gamma = 3$  was the upper boundary of the predefined development grid. Therefore,  $\gamma = 3 \neq$  proven global optimum.

The appropriate interpretation is that the strength of empirical reliability influence appears to affect observed ranking behaviour, making it a worthwhile target for future systematic investigation. A wider development grid, nested validation or evaluation across additional datasets would be required before drawing stronger conclusions about the preferred value of  $\gamma$ .

## 5.5 Practical Implications

The findings suggest that RDRA may be especially useful in deployment-oriented model-selection settings where several conditions are present simultaneously:

- multiple deployment criteria are available
- those criteria produce conflicting candidate rankings
- repeated experimental evidence is available
- criterion rankings differ in reproducibility
- transparent and auditable consensus formation is important

In such settings, RDRA offers an evidence-aware alternative to automatically assigning equal influence to every deployment criterion.

One practical advantage is traceability. Because RDRA retains criterion-specific ranks, stability estimates, reliability weights and Borda contributions, the final consensus can be examined in terms of the evidence that produced it. This may make the model-selection process easier to justify than a single opaque composite score.

The VQ1 and sensitivity results also indicate that RDRA may be particularly useful for producing a stable candidate shortlist rather than asserting an immutable single winner. The Top-3 results suggest that the method can preserve a defensible high-quality candidate set even when uncertainty affects exact candidate ordering.

However, empirical reliability weighting should not replace stakeholder preference analysis. A criterion can be highly reproducible while having low importance in a particular operational context, or less reproducible while remaining essential to a specific stakeholder.

A practical decision process could therefore separate two stages:

1. Stage 1: Empirical reliability-aware shortlist construction
2. Stage 2: Stakeholder-specific final selection

In Stage 1, RDRA uses repeated evidence to determine how confidently criterion-specific rankings should contribute to the deployment consensus. In Stage 2, decision-makers can apply context-specific priorities, organisational constraints and risk preferences to select among the resulting candidates.

This interpretation is consistent with the neutral VQ3 result: empirical reliability and stakeholder preference remain complementary but distinct decision layers.

## 5.6 Threats to Validity and Analytical Limitations

The findings should be interpreted within the limitations of the implemented experimental design.

### **5.6.1 Internal Validity**

Repeated experimental runs, matched comparisons and fixed random states improve comparability, but the estimated reliability values remain conditional on the implemented candidate set, preprocessing procedures, selected deployment metrics and experimental protocol.

Thus, the reliability weights should be interpreted as protocol-dependent empirical estimates rather than intrinsic universal properties of the deployment criteria.

Changes to the candidate models, data-processing pipeline or evaluation metrics could alter criterion-specific reproducibility and therefore alter the resulting RDRA weights.

### **5.6.2 Construct Validity**

The five deployment dimensions, Predictive Quality, Robustness, Probability Reliability, Computational Efficiency and Operational Impact, represent important deployment concerns, but they do not capture every requirement that may matter in a real industrial predictive-maintenance application.

Other settings may additionally require consideration of:

- energy consumption
- interpretability
- cybersecurity
- maintainability
- regulatory compliance
- hardware constraints
- integration cost

The Operational Impact dimension is also based on the implemented asymmetric false-negative/false-positive cost formulation.

### **5.6.3 Statistical Validity**

VQ1 contains 25 genuine criterion-subset perturbations because this number is combinatorially determined by the five deployment criteria.

The use of paired analysis, confidence intervals and Holm correction reduces the risk of overstating multiple comparisons, but the number of available criterion perturbations still constrains inferential precision.

VQ2 contains 16 controlled corruption scenarios. The matched bootstrap with 10,000 resamples provides direct evidence for the responsiveness difference, but a larger set of corruption families,

severities and repeated corruption realisations would increase statistical and empirical confidence in the VQ2 inference.

The VQ3 result should also be interpreted carefully. The absence of an additional demonstrated RDRA advantage is not evidence of formal equivalence because no equivalence or non-inferiority test was conducted.

#### **5.6.4 External Validity**

AI4I provides the primary complete RDRA development and validation environment. Although C-MAPSS extends the experimental context to sequential Remaining Useful Life prediction, it does not provide the same repeated five-dimension evidence structure required for a second independent RDRA reliability-weight estimation. The implementation explicitly maintains this distinction.

Consequently, C-MAPSS provides complementary cross-task deployment-rank responsiveness evidence. Broader generalisation would require additional predictive-maintenance datasets, candidate-model families and ideally real industrial deployment evidence.

#### **5.6.5 Exploratory-Refinement Limitation**

The exploratory refinement selected  $\gamma = 3$  using development-only evidence and locked the value before subsequent VQ2 and VQ3 interpretation.

This procedure reduces validation leakage, but the selected value lies at the upper boundary of the predefined search grid.

Therefore,  $\gamma = 3$  is not a proven globally optimal reliability-influence setting.

The VQ2 result should consequently remain attributed specifically to the locked exploratory refined RDRA, while the primary proposed method remains  $\gamma = 1$ .

Future work could investigate a wider  $\gamma$  range, nested model-selection procedures or dataset-specific reliability-influence calibration without changing the definition of the primary RDRA formulation.

### **5.7 Integrated Discussion**

Therefore, the findings provide support for the central motivation of the research: deployment criteria can provide not only conflicting rankings, but also different levels of empirical support for those rankings.

RDRA introduces an explicit mechanism for incorporating this difference into consensus formation. The VQ1 results indicate that primary RDRA can improve several measures of consistency when criterion availability changes. The sensitivity analysis indicates that the broader

primary consensus is robust to moderate uncertainty in the estimated reliability weights. The VQ2 results further show that the locked exploratory refinement can demonstrate stronger evidence responsiveness than Equal Borda when data quality deteriorates meaningfully.

At the same time, the VQ3 result establishes an important limitation. Improved ranking consistency and responsiveness do not automatically produce superior downstream Top-3 decision usefulness under every stakeholder-preference configuration.

Therefore, the evidence supports the conclusion that reliability-aware rank aggregation provides measurable value for ranking properties and uncertainty conditions, while downstream stakeholder usefulness remains a distinct decision dimension.

## **6. Overall Research Findings**

The analysis provides three principal findings.

First, criterion rankings differ substantially in empirical reproducibility, supporting the central premise that equal aggregation influence does not necessarily reflect equal evidence confidence.

Second, primary RDRA provides statistically supported improvements on several measures of ranking and shortlist consistency under criterion uncertainty, while remaining highly stable under moderate perturbation of its empirical reliability weights.

Third, the locked exploratory refined RDRA demonstrates statistically stronger responsiveness to AI4I data-quality deterioration than Equal Borda, while VQ3 does not demonstrate an additional Top-3 downstream preference advantage.

The evidence therefore supports a bounded research conclusion: Reliability-aware rank aggregation provides measurable value under particular forms of deployment uncertainty.

The strongest evidence concerns criterion uncertainty and data-quality responsiveness rather than universal downstream superiority.

## **7. Conclusion**

This Research Analysis examined the empirical evidence generated by the implemented RDRA framework and evaluated the relationships between criterion reproducibility, ranking consistency, data-quality responsiveness and downstream preference usefulness.

The analysis shows that the five deployment criteria provide unequal levels of ranking reproducibility and that incorporating this information into aggregation can produce measurable benefits. Primary RDRA demonstrates stronger consistency than Equal Borda on several VQ1 measures, while the locked exploratory refinement demonstrates stronger responsiveness under the tested AI4I corruption scenarios. At the same time, the VQ3 analysis does not demonstrate

an additional Top-3 downstream decision-usefulness advantage, establishing an important boundary on the contribution.

Therefore, the findings support RDRA as a reliability-aware and empirically interpretable approach to deployment-oriented predictive-maintenance model selection, while also showing that reliability-aware ranking should complement rather than replace downstream stakeholder decision-making.

## References

- [1] A. H. Alaoui, A. Meddaoui, and M. Hain, “AI-driven predictive maintenance for Industry 4.0: A systematic review of models, methods, and challenges,” *The International Journal of Advanced Manufacturing Technology*, vol. 143, pp. 4861–4876, 2026, doi: 10.1007/s00170-026-17531-w.
- [2] T. Zonta, C. A. da Costa, R. da Rosa Righi, M. J. de Lima, E. S. da Trindade, and G. P. Li, “Predictive maintenance in the Industry 4.0: A systematic literature review,” *Computers & Industrial Engineering*, vol. 150, Art. no. 106889, 2020, doi: 10.1016/j.cie.2020.106889.
- [3] A. Theissler, J. Pérez-Velázquez, M. Kettelgerdes, and G. Elger, “Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry,” *Reliability Engineering & System Safety*, vol. 215, Art. no. 107864, 2021, doi: 10.1016/j.ress.2021.107864.
- [4] O. Serradilla, E. Zugasti, J. Rodriguez, and U. Zurutuza, “Deep learning models for predictive maintenance: A survey, comparison, challenges and prospects,” *Applied Intelligence*, vol. 52, no. 10, pp. 10934–10964, 2022, doi: 10.1007/s10489-021-03004-y.
- [5] H. Engbers and M. Freitag, “Automated algorithm selection for predictive maintenance: Advances and challenges,” *Journal of Manufacturing Systems*, vol. 82, pp. 964–975, 2025, doi: 10.1016/j.jmsy.2025.06.023.
- [6] B. Taşçı, A. Omar, and S. Ayvaz, “Remaining useful lifetime prediction for predictive maintenance in manufacturing,” *Computers & Industrial Engineering*, vol. 184, Art. no. 109566, 2023, doi: 10.1016/j.cie.2023.109566.
- [7] X. Bouthillier et al., “Accounting for variance in machine learning benchmarks,” *Proceedings of Machine Learning and Systems*, vol. 3, pp. 747–769, 2021.
- [8] S. Matzka, “AI4I 2020 Predictive Maintenance Dataset,” *UCI Machine Learning Repository*, 2020, doi: 10.24432/C5HS5C.
- [9] H. Wang, W. Luo, Z. Liu, J. Zhang, and M. Zuo, “Fourier feature refiner network with soft thresholding for machinery fault diagnosis under highly noisy conditions,” *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 22880–22891, 2024, doi: 10.1109/IIOT.2024.3363216.

- [10] T. Silva Filho, H. Song, M. Perello-Nieto, R. Santos-Rodriguez, M. Kull, and P. Flach, "Classifier calibration: A survey on how to assess and improve predicted class probabilities," *Machine Learning*, vol. 112, pp. 3211–3260, 2023, doi: 10.1007/s10994-023-06336-7.
- [11] Y. Lv, X. Guo, Q. Zhou, L. Qian, and J. Liu, "Predictive maintenance decision-making for variable faults with non-equivalent costs of fault severities," *Advanced Engineering Informatics*, vol. 56, Art. no. 102011, 2023, doi: 10.1016/j.aei.2023.102011.
- [12] H. Liao, Y. He, X. Wu, Z. Wu, and R. Bausys, "Reimagining multi-criterion decision making by data-driven methods based on machine learning: A literature review," *Information Fusion*, vol. 100, Art. no. 101970, 2023, doi: 10.1016/j.inffus.2023.101970.
- [13] D. Boix-Cots, F. Pardo-Bosch, and P. Pujadas, "A systematic review on multi-criteria group decision-making methods based on weights: Analysis and classification scheme," *Information Fusion*, vol. 96, pp. 16–36, 2023, doi: 10.1016/j.inffus.2023.03.004.
- [14] A. Tornede, L. Gehring, T. Tornede, M. Wever, and E. Hüllermeier, "Algorithm selection on a meta level," *Machine Learning*, vol. 112, no. 4, pp. 1253–1286, 2023, doi: 10.1007/s10994-022-06161-4.
- [15] M. Mohammadi and J. Rezaei, "Ensemble ranking: Aggregation of rankings produced by different multi-criteria decision-making methods," *Omega*, vol. 96, Art. no. 102254, 2020, doi: 10.1016/j.omega.2020.102254.
- [16] H. Tyralis and G. Papacharalampous, "A review of predictive uncertainty estimation with machine learning," *Artificial Intelligence Review*, vol. 57, Art. no. 94, 2024, doi: 10.1007/s10462-023-10698-8.
- [17] M. G. Kendall, "A new measure of rank correlation," *Biometrika*, vol. 30, nos. 1–2, pp. 81–93, 1938.
- [18] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979, doi: 10.1214/aos/1176344552.
- [19] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [20] A. Saxena and K. Goebel, "Turbofan Engine Degradation Simulation Data Set," *NASA Prognostics Data Repository*, NASA Ames Research Center, Moffett Field, CA, USA, 2008.
- [21] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in *2008 International Conference on Prognostics and Health Management*, Denver, CO, USA, 2008, pp. 1–9, doi: 10.1109/PHM.2008.4711414.